

SOFTWARE REQUIREMENTS SPECIFICATION

Ingredient-Based Meal Planner

A Frontend Web Application for Smart Recipe Discovery
Based on Available Ingredients

Document Type:	Software Requirements Specification (SRS) Document
Prepared For:	College / University Software Engineering Project Submission
Institution:	GLA University
Department:	Department of Computer Science & Engineering
Prepared By:	Aabir Das
Guided By:	Gautum Mukherjee
Date:	19 / 08 / 2026
Standard Followed:	IEEE 830 / IEEE 29148 (adapted)

Confidential — Prepared strictly for academic evaluation purposes

Table of Contents

1. Introduction
 2. Overall Description
 3. System Features and Functional Requirements
 4. External Interface Requirements
 5. Non-Functional Requirements
 6. System Architecture and Design Overview
 7. Data Requirements
 8. Use Case Model
 9. Testing Requirements
 10. Other Requirements
- Appendix A — Glossary

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the Ingredient-Based Meal Planner, a frontend web application designed to help users discover recipes they can prepare using ingredients already available in their kitchen.

1.2 Document Conventions

Requirements in this document are uniquely identified using the prefix pattern **FR-XX** for Functional Requirements and **NFR-XX** for Non-Functional Requirements. The keywords **shall** and **must** indicate mandatory requirements, while **should** indicates optional requirements.

1.3 Project Scope

In Scope (Current Phase): Ingredient selection UI, recipe filtering logic, recipe browsing and detail view, search and category filters, "favorites" saved locally in the browser, missing-ingredient indicator, fully responsive design.

Out of Scope (Current Phase): User login/registration with a live server, real-time multi-user data sync, nutrition tracking with medical accuracy, payment/e-commerce, mobile native apps, AI-generated recipes.

2. Overall Description

2.1 Product Perspective

The Ingredient-Based Meal Planner is a new, standalone frontend product built as a Single Page Application (SPA) that runs entirely in the user's web browser. The recipe dataset is bundled with the application as structured local data (JSON) rather than fetched from a live backend server.

2.2 Product Features (Summary)

- Interactive ingredient picker allowing users to mark ingredients as available or not available
- Smart recipe filtering engine that ranks/filters recipes by ingredient compatibility
- Recipe cards showing title, image, cooking time, difficulty, and match percentage
- Detailed recipe view with full ingredient list, missing-ingredient highlights, and preparation steps
- Search bar and category filters (e.g. Breakfast, Vegetarian, Quick Meals)
- "Favorites" list stored in browser local storage
- Fully responsive design for desktop, tablet, and mobile browsers

2.3 Operating Environment

Client Side: Modern web browsers — Google Chrome, Mozilla Firefox, Microsoft Edge, Safari (latest two major versions).

Operating System: Any OS capable of running a modern browser — Windows, macOS, Linux, Android, iOS.

Screen Sizes: Responsive from 320px (mobile) to 1920px+ (desktop) viewport widths.

3. System Features and Functional Requirements

3.1 Ingredient Selection Module

Description: Allows the user to build a profile of ingredients they currently have, which drives the recipe filtering engine.

ID	Requirement	Priority
FR-01	The system shall display a categorized list of ingredients (e.g. Vegetables, Grains, Dairy, Spices, Proteins) for user selection.	High
FR-02	The system shall allow the user to mark/unmark an ingredient as "available" by clicking/tapping it.	High
FR-03	The system shall provide a search box within the ingredient list to quickly locate an ingredient by name.	Medium
FR-04	The system shall display a live count/summary of the number of ingredients currently selected.	Medium

3.2 Recipe Filtering & Matching Engine

Description: Core business logic of the application. Filters and ranks the recipe dataset based on the user's selected ingredients.

ID	Requirement	Priority
FR-06	The system shall compare each recipe's required-ingredient list against the user's selected ingredients and compute a "match percentage".	High
FR-07	The system shall exclude or visually de-prioritize recipes below a configurable minimum match threshold (e.g. below 50% match).	High
FR-08	The system shall re-calculate and re-render the filtered recipe list in real time whenever the ingredient selection changes.	High
FR-09	The system shall clearly indicate, for each recipe, which required ingredients are missing from the user's selection.	High

4. External Interface Requirements

4.1 User Interfaces

Home / Ingredient Selection Screen: Grid or list of ingredient chips grouped by category, with a persistent "Find Recipes" call-to-action.

Recipe Results Screen: Grid of recipe cards with match percentage badges, filter/sort bar at the top.

Recipe Detail Screen: Full recipe information with a clearly highlighted "Missing Ingredients" panel.

Favorites Screen: List/grid of saved recipes.

Navigation: A persistent top navigation bar with app logo, and on mobile, a collapsible menu.

Color Accent: All primary interactive elements use the food-themed brown accent color (#6B5344) with cream (#F5F1E8) backgrounds to reinforce brand consistency and maintain WCAG AA accessibility standards.

4.2 Software Interfaces

Interface	Purpose
Web Browser	Runtime environment for the SPA (Chrome, Firefox, Edge, Safari)
Local JSON Dataset	Provides recipe and ingredient data in the current phase
Browser LocalStorage API	Persists user favorites and ingredient selection between sessions

5. Non-Functional Requirements

ID	Requirement
NFR-01	The recipe filtering engine shall return updated results within 300ms of an ingredient selection change, for a dataset of up to 200 recipes.
NFR-02	Initial page load shall complete within 3 seconds on a standard broadband connection (≥ 10 Mbps).
NFR-03	A first-time user shall be able to select ingredients and view a filtered recipe list within 3 clicks/taps, without instructions.
NFR-04	The interface shall provide clear visual feedback (e.g. highlight, checkmark) when an ingredient is selected.
NFR-05	The application shall not crash or show a blank screen when zero ingredients are selected; it shall display a helpful empty state.
NFR-06	The application shall run without modification on any modern, standards-compliant web browser across Windows, macOS, Linux, Android, and iOS.

6. System Architecture and Design Overview

6.1 Technology Stack

Layer	Technology	Notes
Frontend Framework	React.js (with Hooks)	Component-based; aligns with future MERN stack
Styling	CSS3 / Tailwind CSS	Food-themed palette, responsive grid/flexbox
State Management	React Context API / useState	Sufficient for this scope
Data (current phase)	Local JSON files	Bundled with the app; no server round-trip
Persistence	Browser LocalStorage API	Favorites, last-used ingredient selection

6.2 Component / Module Breakdown

App Shell — routing/navigation between screens.

IngredientPicker — renders ingredient categories and selection state.

FilterEngine — pure logic module computing match % per recipe (unit-testable).

RecipeCard / RecipeGrid — presentational components for the results list.

RecipeDetail — full recipe view with missing-ingredient highlighting.

FavoritesManager — reads/writes favorites to Local Storage.

SearchAndFilterBar — search box + category chips + sort dropdown.

7. Data Requirements

7.1 Data Entities (Local Dataset)

Ingredient Entity:

Field	Type	Description
id	String	Unique identifier
name	String	Ingredient name (e.g. 'Tomato')
category	String	e.g. Vegetable, Dairy, Spice, Grain, Protein

Recipe Entity:

Field	Type	Description
id	String	Unique identifier
title	String	Recipe name
image	String (URL)	Recipe image reference
category	String	e.g. Breakfast, Dinner, Dessert
cookingTimeMinutes	Number	Preparation + cooking time
difficulty	String	Easy / Medium / Hard
servings	Number	Number of servings
ingredients	Array	List of ingredient IDs + quantities
steps	Array	Ordered preparation instructions

8. Use Case Model

8.1 Primary Use Cases

Use Case	Actor	Description
UC-01: Select Ingredients	User	User browses and selects available ingredients from categorized list
UC-02: View Filtered Recipes	User	System displays recipes ranked by match percentage
UC-03: View Recipe Detail	User	User opens a recipe card to see full details and missing ingredients
UC-04: Mark Recipe as Favorite	User	User saves recipe to favorites stored in local browser storage
UC-05: Search by Name	User	User finds recipes using text search on recipe titles
UC-06: Filter by Category	User	User filters recipes by meal type, dietary preference, or time to cook

9. Testing Requirements

9.1 Testing Scope

Testing shall cover unit-level verification of the filtering/matching logic (core algorithm), component-level UI testing, and manual end-to-end testing across target browsers and screen sizes.

9.2 Test Case Summary

Test ID	Scenario	Expected Result
TC-01	Select 3 ingredients that exactly match a recipe's full ingredient list	Recipe shows 100% match
TC-02	Select 0 ingredients	Empty state message shown, no crash
TC-03	Select ingredients matching only half of a recipe's requirements	Recipe shows ~50% match and lists missing ingredients
TC-04	Search for a recipe by partial name	Matching recipes appear in real time
TC-05	Mark a recipe favorite, then refresh the page	Recipe remains marked as favorite after reload
TC-06	Apply a category filter (e.g. "Breakfast")	Only recipes of that category are shown
TC-07	Resize browser to mobile width (375px)	Layout reflows correctly, no horizontal scroll

10. Other Requirements

10.1 Legal & Compliance

All recipe content and images used in the dataset should be either originally created by the team, sourced from royalty-free/open datasets, or properly attributed, to avoid copyright issues in academic submission and any public deployment.

10.2 Project Assumptions

- The project is evaluated as a frontend-only deliverable unless the curriculum requires a backend phase.
- Team size, timeline, and grading rubric are assumed consistent with a standard final-year/semester project.
- The bundled recipe dataset will contain 40–100+ recipes for a comprehensive demo.
- Users interact with the app in a single session; data persists only via browser local storage.

Appendix A — Glossary

Term	Definition
SPA	Single Page Application — a web app that loads a single HTML page and dynamically updates content
Match Percentage	The proportion of a recipe's required ingredients that the user has selected as available
Local Storage	A browser API used to persist small amounts of data on the user's device between sessions
MERN Stack	MongoDB, Express.js, React.js, Node.js — a JavaScript full-stack combination
FR / NFR	Functional Requirement / Non-Functional Requirement
IEEE 830	IEEE Recommended Practice for Software Requirements Specifications

End of Document